

COL774 Assignment 1 — Report

1. Created Features

Part (c) — Heart rate from BVP/accelerometer/EDA.

Feature	Full name	Modality	Explanation
<code>acf_lag{20..85}</code>	Pulse Self-Similarity Curve (66)	BVP (wrist pulse)	How closely the pulse wave matches a delayed copy of itself, checked at every delay a plausible heart rate could produce. This directly reveals the true beat rate.
<code>acf_peak_*</code>	Best-Match Summary (3)	BVP (wrist pulse)	The strongest match point on that curve, and the heart rate it points to.
<code>bvp_mean...zc</code>	Pulse Wave Shape (8)	BVP (wrist pulse)	How strong, spread out, and jagged the raw pulse wave looks.
<code>acc_*</code>	Wrist Motion (10)	Accelerometer	How much, and in what pattern, the wrist is moving – movement can distort the pulse reading.
<code>eda_mean...slope</code>	Sweat Response (4)	EDA (skin)	Overall sweat-response level, how much it varies, and its trend.

Part (d) — CGM glucose from Empatica E4 + Zephyr BioHarness. Final model: LassoCV, 5-fold CV over a 30-point log-spaced α grid, features standardized before fitting.

Feature	Full name	Modality	Explanation
<code>bvp_*</code>	Pulse Wave Shape & Timing (17)	BVP (wrist pulse)	What the pulse wave looks like and how steady the timing between heartbeats is.
<code>e4_hr_*</code>	Device Heart Rate (7)	Heart rate (wrist sensor)	The wristband's own heart-rate reading, and whether it drifts during the window.
<code>eda_*</code>	Sweat Response (19)	EDA (skin)	How much someone is sweating, and what sudden “stress spike” events look like.
<code>temp_*</code>	Skin Temperature (8)	Temperature	How skin temperature shifts within the window (each person's own baseline removed).
<code>e4_acc_sq_*, bvp_motion_quality</code>	Wrist Motion & Signal Quality (6)	Accelerometer (wrist)	How much the wrist is moving, and whether that motion is corrupting the pulse reading.
<code>ecg_mean...sampen</code>	Heartbeat Waveform & Rhythm (18)	ECG (chest strap)	The heartbeat's electrical shape and how its timing varies – linked to the body's stress/calm balance.
<code>ecg_qt_*, qtc_*, t_*</code>	Heart's Electrical Recovery Time (6)	ECG (chest strap)	How long the heart takes to “reset” after each beat – known to lengthen during low blood sugar.
<code>ptt_mean, ptt_std</code>	Pulse Travel Time (2)	ECG + BVP together	How long the pulse takes to travel from the heart to the wrist.
<code>zephyr_acc_sq_*, ecg_motion_quality</code>	Chest Motion & Signal Quality (6)	Accelerometer (chest)	How much the chest strap is moving, and whether that's corrupting the ECG reading.
<code>breathing_*</code>	Breathing Pattern (13)	Breathing (chest strap)	How breathing depth, rate, and trend change within the window.

2. Normalized Acceleration

Figure 1 shows a representative Part (a) window (BVP, $\tilde{a}_{sq}(t)$, EDA, target HR); Figure 2 shows a representative Part (d) window (BVP, $\tilde{a}_{sq}(t)$, EDA, ECG, on-device heart rate, target CGM glucose). In the Part (d) window, two motion episodes (visible as spikes in $\tilde{a}_{sq}(t)$ around $t=40\text{--}110\text{s}$) coincide with BVP/EDA artifact and a heart-rate rise – exactly the motion-corruption effect the signal-quality ratio features are designed to flag.

Part (a) representative window -- target heart rate = 83.4 bpm

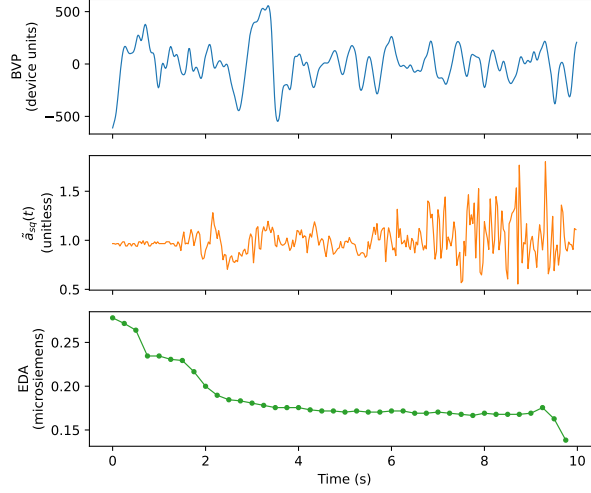


Figure 1: Representative Part (a) window.

3. Top Five Features

Importance is $|\text{standardized regression coefficient}|$ (features standardized before comparison), per PDF instructions.

Part (c) (Ridge, $\alpha=0.01$):

Feature	Modality	std. coef.	Why relevant
acf_lag28	BVP (autocorrelation)	101.13	Match strength at the delay corresponding to ≈ 137 bpm – close to this window’s actual heart rate.
acf_lag50	BVP (autocorrelation)	98.34	Match strength at the ≈ 77 bpm delay.
acf_lag26	BVP (autocorrelation)	89.64	Match strength at the ≈ 148 bpm delay.
acf_lag48	BVP (autocorrelation)	78.92	Match strength at the ≈ 80 bpm delay.
acf_lag30	BVP (autocorrelation)	78.79	Match strength at the ≈ 128 bpm delay.

Several adjacent-lag ACF values receiving large, opposite-sign coefficients lets the linear model interpolate the true period at sub-sample resolution, rather than relying on one discrete peak lag.

Part (d) (LassoCV; protocol d3, cross-subject – the hardest, most heavily-weighted split):

Feature	Modality	std. coef.	Why relevant
ecg_qt_mean	ECG (Zephyr)	7.38	How long the heart takes to “reset” after each beat – this stretches out during low blood sugar in Type 1 diabetes.
ecg_min	ECG (Zephyr)	1.57	The lowest point of the heartbeat’s electrical waveform (each person’s own baseline removed).
breathing_std	Breathing (Zephyr)	1.18	How much breathing depth varies within the window (baseline removed).
ecg_lf_hf_ratio	ECG HRV (Zephyr)	0.74	Balance between the body’s “alert” and “calm” nervous-system activity; expected to shift during a glucose swing.
ecg_sd1_sd2_ratio*	ECG HRV (Zephyr)	0.37	A geometric comparison of short-term vs. longer-term heartbeat-timing variability.

*Lasso keeps only 4 features nonzero in the d3 fit (the four above it); the 5th is the strongest signal from the d2 fit – overall, importance

Part (d) representative window -- target CGM glucose = 138 mg/dL

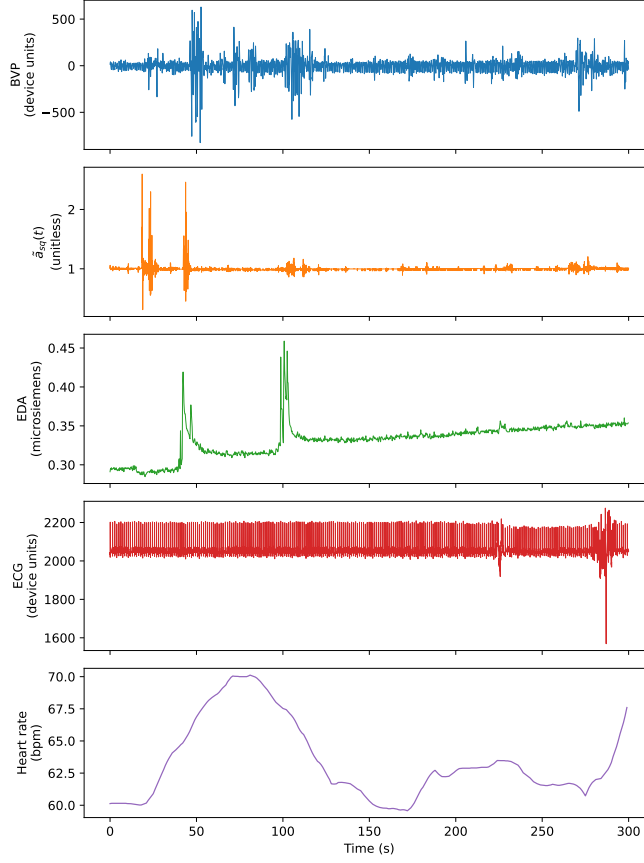


Figure 2: Representative Part (d) window.

concentrates almost entirely in ECG timing/HRV and breathing.

4. Median Baseline Comparison

Part / Protocol	Model		Median baseline	
	NMAE	NMSE	NMAE	NMSE
Part (c) (local val., $n=31,797$)	0.6434	0.5002	0.9918	1.0141
Part (d), d1 (random within-subject)	0.9277	0.9056	0.9709	1.0624
Part (d), d2 (temporal within-subject)	0.9328	0.9592	0.9951	1.1429
Part (d), d3 (cross-subject)	1.0038	1.0935	1.0958	1.3660